

Multimodal Speech Emotion Recognition via Parameter-Efficient Fine-Tuning of Audio-Language Models

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Abstract

Speech emotion recognition remains challenging due to strong variability in speaking style, dataset characteristics, and the unreliable contribution of linguistic information across domains. This work aims to evaluate whether instruction-following audio–language models can be effectively adapted for speech emotion recognition and to analyze the role of parameter-efficient fine-tuning and multimodal inputs. We formulate emotion recognition as a constrained text generation task and apply LoRA and DoRA parameter-efficient fine-tuning to adapt a pretrained audio–language model while keeping the backbone frozen. We conduct experiments on two benchmark datasets with contrasting properties, namely the acted Emotional Speech Dataset (ESD) and the conversational IEMOCAP dataset, under both audio-only and audio-plus-transcript settings. Results show that in the zero-shot setting, balanced performance is limited, with best macro-averaged F1 scores of 0.37 on IEMOCAP and 0.13 on ESD. After parameter-efficient fine-tuning, performance improves substantially, achieving a macro F1 of 0.63 on IEMOCAP with DoRA and 0.84 on ESD with LoRA, confirming that lightweight adaptation methods are critical for achieving reliable emotion recognition in instruction-following audio–language models.

Keywords

Speech Emotion Recognition, Audio-Language Models, Parameter-Efficient Fine-Tuning, Multimodal Learning, Low-Rank Adaptation

1 Introduction

Speech emotion recognition (SER) aims to infer a speaker’s emotional state from speech signals and plays an important role in applications such as human–computer interaction, assistive technologies, and behavioral analysis [9]. Despite extensive research, SER remains a challenging task. Emotional expression varies widely across speakers, recording conditions, speaking styles, and datasets, and models trained under supervised settings often struggle to generalize beyond the specific conditions observed during training.

Recent advances in large-scale foundation models have introduced instruction-following audio–language models capable of performing a wide range of speech understanding tasks through natural-language prompting [7]. Rather than relying on task-specific classification heads, these models generate textual outputs conditioned on audio inputs and instructions, enabling zero-shot inference on previously unseen tasks.

In this work, we investigate the application of an instruction-following audio–language model to speech emotion recognition by formulating SER as a constrained text generation task. Given an input utterance and a task instruction prompt (audio-only or

audio+transcript), the model is required to output exactly one emotion label from a predefined set. This approach allows SER to be performed without modifying the model architecture or introducing additional classification layers. However, it remains unclear whether such models can reliably perform emotion recognition under class imbalance and domain shifts using prompting alone.

To address these limitations, we explore parameter-efficient fine-tuning (PEFT) methods that adapt large pretrained models to the SER task while keeping the backbone frozen. In particular, we study Low-Rank Adaptation (LoRA) and Weight-Decomposed Low-Rank Adaptation (DoRA), which introduce a small number of trainable parameters and significantly reduce training cost compared to full fine-tuning. These methods provide a practical mechanism to specialize instruction-following models for emotion recognition without sacrificing their general-purpose representations.

We further examine the role of multimodality in SER by comparing audio-only inference with a multimodal setting that incorporates textual transcripts. While linguistic content can provide useful contextual information in conversational speech, transcripts may be ambiguous or weakly informative in acted datasets. Understanding when textual information supports emotion recognition and when it introduces noise is therefore essential for robust model design.

We evaluate our approach on two benchmark datasets with complementary characteristics: the acted ESD dataset [5] and the conversational IEMOCAP dataset [4]. Based on these settings, we address the following research questions:

- **RQ1:** How effective is zero-shot speech emotion recognition using an instruction-following audio–language model without task-specific training?
- **RQ2:** How does parameter-efficient fine-tuning affect emotion recognition performance, and how do LoRA and DoRA compare under identical conditions?
- **RQ3:** What is the impact of incorporating textual transcripts as an additional modality across different speech domains?

The remainder of this paper is organized as follows: we review related work, describe the proposed formulation and adaptation approach, and then present the experimental setup and results, followed by a concluding discussion. All scripts and files are available in the following public GitHub repository: 2025-P7-MER.

2 Related Work

Speech emotion recognition (SER) has been widely studied using both handcrafted acoustic descriptors and deep learning models trained on spectrograms or raw waveforms. Recent reviews

show that although modern architectures can achieve strong results within a single benchmark, performance often drops when the speaking style, speakers, or recording setup changes [6, 9]. This weakness is commonly attributed to dataset-specific biases and differences in annotation protocols, which makes cross-dataset generalization a key open challenge. Motivated by these findings, we evaluate robustness by fine-tuning on acted speech (ESD) and testing on conversational speech (IEMOCAP).

A newer direction is to use instruction-following audio-language models, which treat speech understanding tasks as conditional text generation. Models in this family are trained on paired audio and text and can respond to natural-language prompts, enabling zero-shot task transfer without adding a separate classifier head [7, 10]. However, several works suggest that prompting alone is often unreliable for fine-grained recognition problems, especially under class imbalance or domain shift, where models may collapse to dominant labels. Based on this paradigm, we formulate SER as constrained generation and explicitly study the limits of zero-shot prompting.

To adapt large pretrained models efficiently, parameter-efficient fine-tuning (PEFT) methods update a small set of additional parameters while keeping the backbone frozen. LoRA inserts low-rank updates into selected weight matrices, while DoRA further stabilizes learning by decoupling weight magnitude from direction, which can be helpful in low-rank settings [2, 3]. In parallel, QLoRA reduces the memory cost of adaptation by quantizing the frozen weights and training only the lightweight adapters [8]. In our work, we combine this PEFT setup with an instruction-based SER formulation and compare LoRA and DoRA under identical ranks and modalities.

3 Methodology

3.1 Problem Formulation

We address the task of categorical speech emotion recognition (SER), where the objective is to assign a single emotion label to a spoken utterance. In this work, we consider a four-class classification setting with the following emotion set:

$$\mathcal{E} = \{\text{Angry, Happy, Sad, Neutral}\}.$$

Let a denote an audio signal corresponding to a speech utterance, and let t denote its textual transcription when available. Depending on the experimental configuration, the model input is either audio-only a or multimodal (a, t) . Given an input $x \in \{a, (a, t)\}$, the goal is to predict an emotion label $e \in \mathcal{E}$.

Rather than formulating SER as a conventional classification task with a task-specific output layer, we cast it as a conditional text generation problem. Given an input x and a natural-language instruction prompt p , the model generates a short textual output corresponding to one of the emotion labels in $\mathcal{E}$. This formulation enables the use of instruction-following audio-language models for SER without modifying their architecture.

Formally, the prediction rule is defined as:

$$\hat{e} = \arg \max_{e \in \mathcal{E}} P(e \mid p, x),$$

where $P(e \mid p, x)$ denotes the conditional probability of generating emotion label e given the input signal and the instruction

prompt. The exact instruction prompts used for the audio-only and audio+transcript settings are reported in Appendix A.

3.2 Model Architecture

We build our SER system on **Voxtral-mini-3B** [1], an instruction-following audio-language model composed of (i) a pretrained speech encoder, (ii) an *audio-language adapter* that downsamples audio embeddings, and (iii) a pretrained decoder-only language model that generates text autoregressively. Following the Voxtral design, the speech encoder is based on Whisper large-v3 and produces high frame-rate acoustic embeddings, which are then downsampled by an MLP adapter to obtain a shorter sequence suitable for the language decoder.

Given an utterance waveform a , the audio encoder produces acoustic features $\mathbf{h}_a$, which are mapped by the adapter to projected and downsampled embeddings $\tilde{\mathbf{h}}_a$ in the language model embedding space. When transcripts are available, the text t is tokenized and embedded as $\mathbf{h}_t$. We then construct a single prefix context for the decoder by concatenating the instruction prompt embeddings with the audio embeddings and (optionally) transcript embeddings:

$$s = [\mathbf{e}(p); \tilde{\mathbf{h}}_a; \mathbf{h}_t],$$

where $\mathbf{h}_t$ is omitted in the audio-only setting.

Speech emotion recognition is cast as *constrained text generation*: given a fixed prompt p with a modality-specific variant (Appendix A), the model must output exactly one label from $\mathcal{E}$.

Conditioned on s , the decoder generates an output sequence $y = (y_1, \dots, y_T)$ autoregressively:

$$P(y \mid s) = \prod_{i=1}^T P(y_i \mid y_{<i}, s).$$

At inference time, we decode a short continuation and map it to the closest valid label in $\mathcal{E}$.

3.3 Parameter-Efficient Fine-Tuning

While instruction-based zero-shot inference enables direct application of the audio-language model to speech emotion recognition, it does not provide sufficient performance or robustness across datasets. To adapt the model to the SER task without incurring the computational cost and overfitting risks of full fine-tuning, we employ parameter-efficient fine-tuning (PEFT) methods.

In all fine-tuning experiments, the pretrained audio encoder and audio-language adapter are frozen; in the language model we freeze the base weights and train only LoRA/DoRA adapter parameters. Task-specific adaptation is achieved by introducing a small number of trainable parameters into selected components of the language model decoder. We restrict PEFT to the self-attention projections in each decoder layer ℓ , learning $\Delta W_*^{(\ell)}$ only for $* \in \{q, k, v, o\}$, while all other weights remain frozen.

Low-Rank Adaptation (LoRA). LoRA introduces trainable low-rank updates to selected weight matrices in the transformer decoder. Given a frozen weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA defines the adapted weights as:

$$W' = W + \frac{\alpha}{r} BA,$$

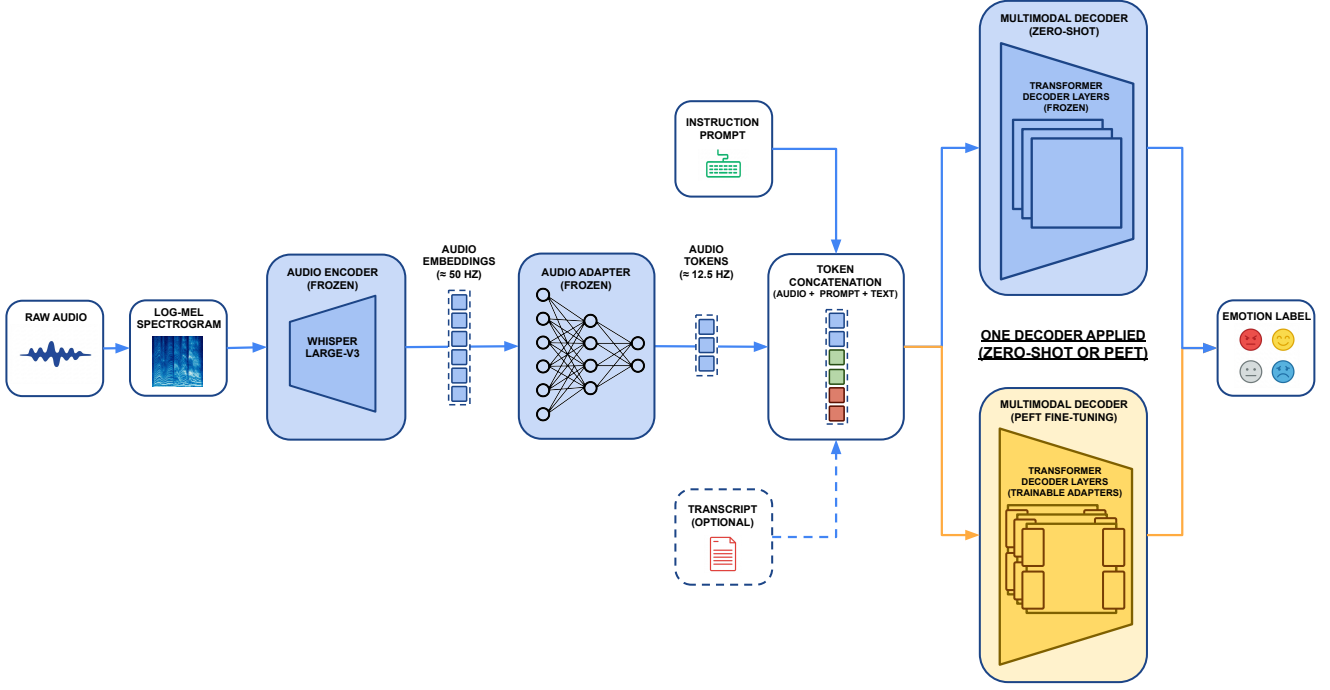


Figure 1: Overview of the pipeline based on Voxtral-mini-3B: a frozen Whisper large-v3 encoder and frozen audio–language adapter produce audio tokens, which are concatenated with the instruction prompt and optional transcript tokens. Decoding is performed either fully frozen (zero-shot) or with LoRA/DoRA adapters, producing one label from {Angry, Happy, Sad, Neutral}.

where $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$, $r \ll \min(d, k)$ is the rank of the adaptation, and α is a scaling constant. During training, only the low-rank matrices A and B are updated, while the original weight matrix W remains frozen [2].

Weight-Decomposed Low-Rank Adaptation (DoRA). DoRA extends LoRA by decoupling the magnitude and direction of the adapted weights to improve optimization stability. Instead of directly updating the full weight matrix, DoRA reparameterizes the adapted weights as:

$$W' = \lambda \cdot \frac{W + \Delta W}{\|W + \Delta W\|_F},$$

where ΔW is defined as in LoRA, $\|\cdot\|_F$ denotes the Frobenius norm, and λ is a learnable scalar controlling the overall weight magnitude. By separating scale and direction, DoRA enables more stable training dynamics in low-rank adaptation settings [3].

4 Experiments

4.1 Datasets

We evaluate our approach on two widely used speech emotion recognition datasets with complementary characteristics: the Emotional Speech Dataset (ESD) and the Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset. These datasets differ substantially in speaking style, recording conditions, and the reliability of

linguistic cues, allowing us to assess model behavior across both acted and conversational speech.

Emotional Speech Dataset. ESD is a large-scale corpus of acted emotional speech recorded under controlled studio conditions. The dataset contains clearly articulated emotional expressions with relatively balanced class distributions. In our experiments, we use a fixed stratified split consisting of **8,000 samples for training**, **400 samples for validation**, and **5,600 samples for testing**. This split is kept consistent across all experiments involving ESD and is used for parameter-efficient fine-tuning as well as evaluation [5]. ESD includes five emotions; however, to match IEMOCAP’s four-class setting and enable cross-domain comparison, we exclude Surprise and use the shared set Angry, Happy, Sad, Neutral.

Interactive Emotional Dyadic Motion Capture. IEMOCAP consists of dyadic conversational speech recordings collected in semi-natural interaction scenarios. Compared to ESD, it exhibits higher variability in speaking style, spontaneous emotional expression, and notable class imbalance. Following common practice, the original emotion annotations are mapped to a four-class categorical setting (Angry, Happy, Sad, Neutral) by merging the *excited* category with *happy*. In this work, IEMOCAP is used **only for evaluation** to assess cross-domain generalization: all PEFT models are fine-tuned on ESD and then evaluated on a fixed IEMOCAP subset of **1,085 samples** [4].

Table 1: Zero-shot performance of Voxtral-mini-3B on ESD and IEMOCAP under audio-only and audio+transcript inputs. Bold indicates the best result per modality within each dataset.

Modality	Class-wise F1				Global Metrics		
	Angry	Happy	Neutral	Sad	Accuracy	Kappa	Macro F1
ESD							
Audio only	0.11	0.05	0.32	0.07	0.20	0.00	0.11
Audio + Transcript	0.11	0.06	0.33	0.11	0.21	0.01	0.13
IEMOCAP							
Audio only	0.39	0.20	0.56	0.23	0.45	0.17	0.34
Audio + Transcript	0.37	0.29	0.57	0.25	0.46	0.20	0.37

For both datasets, raw audio recordings are used as input in all experiments. In the audio-only configuration, predictions are generated solely from the speech signal. In the audio-plus-transcript configuration, ground-truth transcripts provided with the datasets are included as an additional input modality. Transcripts are not used in the audio-only setting.

4.2 Evaluation Metrics

Speech emotion recognition datasets can be imbalanced, so we report metrics that reflect both overall correctness and balanced class-wise behavior. Specifically, we use **Accuracy**, **macro-averaged F1**, and **Cohen’s kappa**. Accuracy measures the fraction of correctly classified utterances, but it can be inflated by majority classes (e.g., *Neutral*). Macro F1 computes the F1 score independently for each emotion class and averages them, giving equal weight to frequent and rare emotions. Cohen’s kappa measures agreement between predictions and ground truth while accounting for chance-level agreement, providing a more reliable view than accuracy alone.

To quantify the impact of transcripts, we additionally report a **Semantic Interference** score, defined as the change in a metric $\mathcal{M}$ when adding text to audio:

$$\Delta_{SI} = \mathcal{M}(a, t) - \mathcal{M}(a),$$

where a denotes the audio-only setting and (a, t) denotes audio plus transcript. In this study, we focus on $\mathcal{M} = \text{Macro F1}$. Positive values indicate that transcripts provide complementary information, while negative values indicate that transcripts act as noise and degrade performance.

4.3 Implementation Details

Zero-shot experiments were run on Google Colab using an NVIDIA T4 GPU (16 GB VRAM). Fine-tuning was performed on a single NVIDIA RTX 2080 Ti (11 GB VRAM) with 64 GB system RAM. To fit the 3B-parameter model on consumer hardware, we used QLoRA with 4-bit quantization while keeping the backbone frozen, and applied the same training and decoding settings across all models and modalities for fair comparison. Due to GPU memory constraints, we trained with batch size 1 and 16-step gradient accumulation

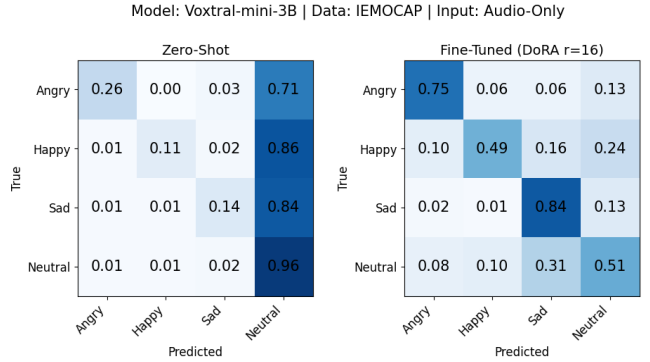


Figure 2: Normalized confusion matrices on IEMOCAP for Voxtral-mini-3B, comparing zero-shot inference and PEFT fine-tuning (DoRA $r=16$). Fine-tuning reduces the strong Neutral-class collapse and improves class-wise separation.

(effective batch size 16) for up to 3 epochs, using early stopping, a learning rate of 10^{-4} , the 8-bit AdamW optimizer, and a fixed seed of 42 for reproducibility.

4.4 Zero-Shot Results

We first evaluate Voxtral-mini-3B in a zero-shot setting, where no task-specific fine-tuning is applied and predictions are generated only through instruction-based prompting. This experiment examines whether emotional information learned during pretraining is sufficient for speech emotion recognition without adaptation. Zero-shot results on IEMOCAP and ESD are reported in Table 1.

On the IEMOCAP dataset, zero-shot performance is limited. Although overall accuracy reaches around 0.45, balanced metrics indicate weak performance. Macro-averaged F1 scores remain low (0.34 for audio-only and 0.37 with transcripts), and Cohen’s kappa values (0.17–0.20) suggest only slight agreement beyond chance. Class-wise F1 scores show a strong bias toward the *Neutral* class, which is recognized relatively well, while emotions such as *Happy*

Table 2: PEFT results on ESD (in-domain) and IEMOCAP (cross-domain, fine-tuned on ESD) in the audio-only setting. Bold indicates the best result per column within each dataset.

Model	Class-wise F1				Global Metrics		
	Angry	Happy	Neutral	Sad	Accuracy	Kappa	Macro F1
ESD (In-Domain)							
LoRA ($r=8, \alpha=16$)	0.84	0.82	0.77	0.67	0.78	0.71	0.78
LoRA ($r=16, \alpha=32$)	0.88	0.85	0.85	0.80	0.85	0.79	0.84
DoRA ($r=8, \alpha=16$)	0.84	0.82	0.78	0.64	0.78	0.71	0.77
DoRA ($r=16, \alpha=32$)	0.85	0.81	0.85	0.78	0.83	0.77	0.82
IEMOCAP (Cross-Domain)							
LoRA ($r=8, \alpha=16$)	0.60	0.06	0.56	0.03	0.45	0.18	0.31
LoRA ($r=16, \alpha=32$)	0.73	0.49	0.55	0.54	0.57	0.42	0.58
DoRA ($r=8, \alpha=16$)	0.73	0.19	0.59	0.20	0.52	0.30	0.43
DoRA ($r=16, \alpha=32$)	0.74	0.59	0.56	0.61	0.62	0.49	0.63

and *Sad* are poorly detected. This behavior is visually confirmed in Figure 2.

Zero-shot performance on the ESD dataset is substantially worse. Macro-averaged F1 scores are close to random (0.11–0.13), and Cohen’s kappa values are approximately zero. The model largely collapses to predicting the *Neutral* class, with very low F1 scores for the remaining emotions.

Overall, instruction-based zero-shot inference is insufficient for reliable SER: predictions are strongly biased and poorly balanced, especially on acted speech. This motivates parameter-efficient fine-tuning, which is analyzed in the following subsection.

4.5 Parameter-Efficient Fine-Tuning Results

Table 2 summarizes **audio-only** PEFT performance on **ESD** (in-domain) and **IEMOCAP** (cross-domain evaluation after fine-tuning on ESD), reporting both class-wise F1 and global metrics.

Cross-domain Results on IEMOCAP. On IEMOCAP, parameter-efficient fine-tuning substantially improves over zero-shot inference across all configurations. The strongest audio-only cross-domain performance is obtained by **DoRA** ($r=16$), reaching **0.63** macro F1 with **0.62** accuracy and **0.49** Cohen’s kappa (Table 2). Under tighter parameter budgets ($r=8$), performance decreases for both methods, but the degradation is much stronger for LoRA, which remains highly unbalanced across classes. DoRA is comparatively more robust at $r=8$, retaining higher macro F1 (0.43) and more stable class-wise scores.

In-domain Results on ESD. On ESD, **LoRA** ($r=16$) achieves the best audio-only performance, with **0.84** macro F1 and **0.85** accuracy (Table 2). In this lower-variability setting, increasing rank yields consistent gains for both LoRA and DoRA, and the simpler LoRA adaptation is sufficient to reach the highest balanced performance.

Beyond the aggregate metrics, the main cross-domain gains come from recovering the harder emotion classes rather than improving the already-strong *Neutral* predictions. For instance, under low capacity LoRA ($r=8$) almost collapses on *Happy* and *Sad* (F1 of 0.06 and 0.03), whereas the best configuration DoRA ($r=16$) yields much more balanced recognition across all four classes (e.g., *Happy* 0.59 and *Sad* 0.61), which is reflected in the higher Macro F1 and Cohen’s kappa.

Overall, parameter-efficient fine-tuning yields a clear performance improvement over zero-shot inference, substantially reducing class bias and improving balanced emotion recognition. The relative effectiveness of adaptation strategies is domain-dependent: DoRA is more robust on spontaneous conversational speech, particularly under constrained parameter budgets, whereas LoRA performs best on acted speech with consistent acoustic patterns. These results emphasize the importance of selecting adaptation methods in accordance with the characteristics of the target dataset.

4.6 Input Modality Analysis

The impact of adding transcripts is summarized using the Semantic Interference score Δ_{SI} (Table 3), computed as the change in macro-averaged F1 when moving from audio-only to audio+transcript inputs.

On **ESD (in-domain)**, Δ_{SI} is non-positive for all configurations, indicating that transcripts do not reliably help acted emotion recognition. A representative case is **LoRA** ($r=8$), where adding transcripts reduces macro F1 from **0.78** to **0.53** ($\Delta_{SI}=-0.25$). Illustratively, acted speech often uses semantically neutral or reusable sentences (e.g., everyday statements) that can be performed with different emotional prosody; in such cases the transcript provides limited class information and can compete with strong acoustic cues.

Table 3: Modality Impact on macro-averaged F1 (Δ_{SI}). Positive values indicate complementary information, while negative values indicate interference. Largest gain is highlighted in green and largest drop in red.

Model	Audio only	Audio + Transcript	Δ_{SI}
ESD (In-Domain)			
LoRA ($r=8, \alpha=16$)	0.78	0.53	-0.25
LoRA ($r=16, \alpha=32$)	0.84	0.84	-0.01
DoRA ($r=8, \alpha=16$)	0.77	0.72	-0.05
DoRA ($r=16, \alpha=32$)	0.82	0.82	0.00
IEMOCAP (Cross-Domain)			
LoRA ($r=8, \alpha=16$)	0.31	0.35	+0.04
LoRA ($r=16, \alpha=32$)	0.58	0.59	+0.01
DoRA ($r=8, \alpha=16$)	0.43	0.52	+0.10
DoRA ($r=16, \alpha=32$)	0.63	0.62	0.00

On **IEMOCAP (cross-domain)**, the effect is mixed and generally smaller. Under constrained capacity, transcripts can provide complementary lexical cues: **DoRA** ($r=8$) improves macro F1 from **0.43 to 0.52** ($\Delta_{SI} = +0.10$). In conversational speech, emotionally informative phrasing (e.g., disagreement, blame, enthusiasm) can add signal beyond prosody, which becomes more valuable when the acoustic adaptation is limited. At higher rank, gains largely vanish (e.g., **DoRA** $r=16$: 0.63 $\rightarrow$ 0.62), suggesting that once acoustic modeling is strong enough, transcripts add little in this setup.

Overall, transcripts behave as a conditional modality: they may help conversational speech under limited adaptation capacity, but can introduce interference in acted speech where lexical content is weakly aligned with emotion. This motivates selective, domain-aware fusion that can downweight transcripts when they are ambiguous rather than concatenating modalities uniformly.

5 Conclusion

This work studied whether an instruction-following audio-language model can be reliably used for speech emotion recognition (SER), and how parameter-efficient fine-tuning (PEFT) and transcript-based multimodality affect performance across speech domains. We cast SER as constrained text generation with a fixed prompt with a modality-specific variant and adapted Voxtral-mini-3B using LoRA and DoRA while keeping the backbone frozen, comparing audio-only and audio-plus-transcript inputs on acted ESD (in-domain) and conversational IEMOCAP (cross-domain).

Our results support three main conclusions. (RQ1) Zero-shot prompting performs poorly, with very low macro F1 on both datasets (0.13 on ESD and 0.37 on IEMOCAP). In contrast, parameter-efficient fine-tuning greatly improves balanced emotion recognition, achieving 0.84 macro F1 on ESD and 0.63 on IEMOCAP. (RQ2) The effectiveness of PEFT methods depends on the domain. For acted speech, LoRA gives the best in-domain results (0.84 macro F1), whereas in cross-domain settings DoRA is more robust, reaching 0.63 macro F1 on IEMOCAP after fine-tuning on ESD. (RQ3) Transcripts function as a conditional modality. On ESD they consistently hurt performance, with the largest drop from 0.78 to 0.53 ($\Delta_{SI} = -0.25$). On IEMOCAP they offer only small improvements and mainly help

when model capacity is limited, with the largest gain from 0.43 to 0.52 ($\Delta_{SI} = +0.10$).

Future work will focus on multimodal fusion mechanisms that adaptively estimate when transcripts are trustworthy, for example through confidence-based weighting or explicit checks for disagreement between acoustic and textual cues. We also plan to extend evaluation to more realistic conditions, including ASR-generated transcripts, multilingual speech, and larger emotion sets. Finally, we will test robustness under missing, corrupted, or mismatched transcripts, since such cases are directly linked to reliability, safety, and privacy-sensitive deployment scenarios.

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A Prompts

Audio-only

You are an emotion classifier for speech.
Possible emotions: Angry, Happy, Sad, Neutral.
From the given audio, classify the SPEAKER's emotion.
Answer with EXACTLY one word from this set: Angry, Happy, Sad, Neutral.
Do not add extra words.

Audio-plus-transcript

You are an emotion classifier for speech.
Possible emotions: Angry, Happy, Sad, Neutral.
From the given audio and its transcript, classify the SPEAKER's emotion.
Answer with EXACTLY one word from this set: Angry, Happy, Sad, Neutral.
Do not add extra words.